

18 A $R = \frac{\rho l}{A}, \quad V = Al$

When re-cast, length = $l/2$, cross-sectional area = $2A$ since V is conserved.

Thus new resistance, $R' = \frac{\rho l / 2}{2A} = \frac{1}{4} \frac{\rho l}{A} = \frac{1}{4} R$

19 C Energy = power x time = $VIt = VO = 12 \times 4.0 = 48. \text{J}$